

EC601 — Control System

Semester VI • 3L:0T:0P • 3 Credits

Module 5 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1	State Variable Analysis — Introduction	2
1.1	Why State Variables?	2
2	State Space Representation	2
2.1	Choosing State Variables	3
2.2	Phase Variable Form — Example	3
3	State Transition Matrix (STM)	4
3.1	Definition and Computation	4
3.2	Properties of the STM	5
4	Solution of State Equations	5
4.1	Homogeneous State Equation (Zero Input)	5
4.2	Non-Homogeneous State Equation (With Input)	6
4.3	Laplace Domain Solution	6
5	Transfer Function from State Space	6
6	Controllability	7
6.1	Kalman's Controllability Test	8
6.2	Physical Interpretation	8
7	Observability	8
7.1	Kalman's Observability Test	9
7.2	Physical Interpretation	9
7.3	Controllability vs Observability — Duality	9

1. State Variable Analysis — Introduction

Classical control (transfer function approach) describes a system only in terms of its input-output relationship and is limited to LTI, SISO systems. **State variable analysis** (modern control) provides a complete internal description of the system, handles MIMO systems, time-varying systems, and nonlinear systems, and is directly suited for computer-based design.

State: The state of a system at time t_0 is the minimum set of variables (state variables) that, together with the input for $t \geq t_0$, completely determines the future behaviour of the system.

State Variable: Each member of the minimum set of variables that completely describes the state of the system. For an n -th order system, there are exactly n state variables.

State Vector: The column vector $\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_n(t)]^T$ formed by all n state variables.

State Space: The n -dimensional space whose axes are the state variables $x_1, x_2, \dots, x_n$. Each point in state space represents a unique state of the system.

1.1 Why State Variables?

Aspect	Transfer Function	State Space
System type	SISO, LTI only	MIMO, nonlinear, time-varying
Internal info	No (black box)	Yes (complete internal description)
Initial conditions	Difficult to handle	Naturally included
Computer design	Indirect	Direct (matrix methods)
Stability analysis	Characteristic equation	Eigenvalues of $\mathbf{A}$ matrix
Design methods	Root locus, Bode	Pole placement, optimal control

2. State Space Representation

Standard State Space Form

State equation:

$$\dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t) + \mathbf{B} \mathbf{u}(t)$$

Output equation:

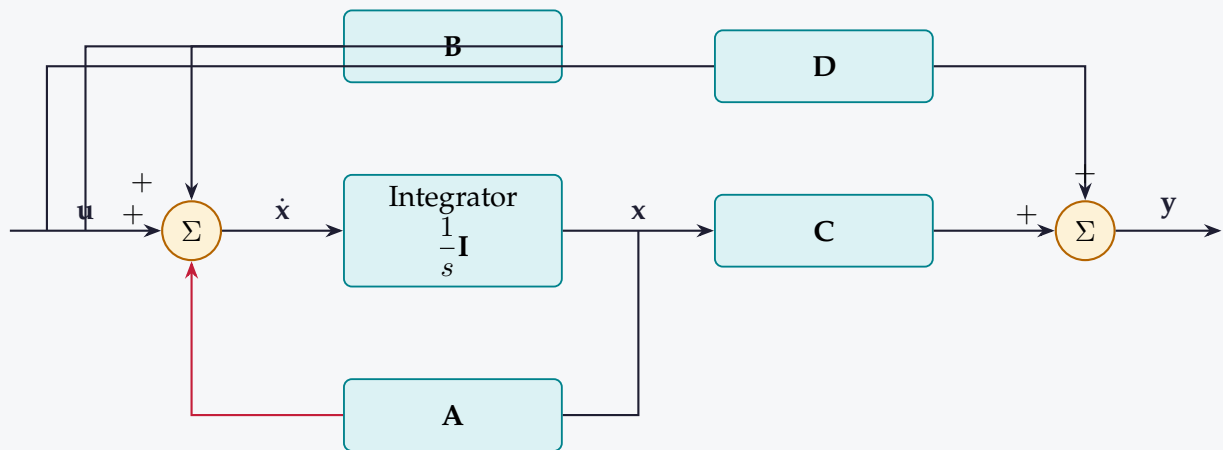
$$\mathbf{y}(t) = \mathbf{C} \mathbf{x}(t) + \mathbf{D} \mathbf{u}(t)$$

where:

- $\mathbf{x}(t) \in \mathbb{R}^n$ — state vector (n state variables)
- $\mathbf{u}(t) \in \mathbb{R}^r$ — input vector (r inputs)

- $\mathbf{y}(t) \in \mathbb{R}^m$ — output vector (m outputs)
- $\mathbf{A}$ ($n \times n$) — system/plant matrix (dynamics)
- $\mathbf{B}$ ($n \times r$) — input matrix
- $\mathbf{C}$ ($m \times n$) — output matrix
- $\mathbf{D}$ ($m \times r$) — direct transmission (feedthrough) matrix

State Space Block Diagram



2.1 Choosing State Variables

For a physical system, state variables are typically the **energy storage variables**:

- **Electrical circuits:** Capacitor voltages v_C and inductor currents i_L
- **Mechanical systems:** Displacement x and velocity $\dot{x}$
- **Thermal systems:** Temperatures at each node

For an n -th order ODE, the standard choice is the **phase variable form**: $x_1 = y$, $x_2 = \dot{y}$, ..., $x_n = y^{(n-1)}$.

2.2 Phase Variable Form — Example

Example: 3rd Order System

Given ODE: $\ddot{y} + a_2\ddot{y} + a_1\dot{y} + a_0y = b_0u$

State variables: $x_1 = y$, $x_2 = \dot{y}$, $x_3 = \ddot{y}$

State equations:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -a_0x_1 - a_1x_2 - a_2x_3 + b_0u$$

Matrix form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ b_0 \end{bmatrix}}_{\mathbf{B}} u, \quad y = \underbrace{\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}}_{\mathbf{C}} \mathbf{x}$$

The $\mathbf{A}$ matrix in this form is called the **companion matrix** or **phase variable canonical form**.

3. State Transition Matrix (STM)

State Transition Matrix (STM) $\Phi(t)$: Also called the **matrix exponential** $e^{\mathbf{A}t}$. It describes how the state vector transitions from its initial value $\mathbf{x}(0)$ to $\mathbf{x}(t)$ in the absence of any input (homogeneous case):

$$\mathbf{x}(t) = \Phi(t) \mathbf{x}(0) = e^{\mathbf{A}t} \mathbf{x}(0)$$

3.1 Definition and Computation

Matrix Exponential — Definition

$$\Phi(t) = e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2 t^2}{2!} + \frac{\mathbf{A}^3 t^3}{3!} + \dots = \sum_{k=0}^{\infty} \frac{(\mathbf{A}t)^k}{k!}$$

Methods to compute $e^{\mathbf{A}t}$:

1. **Infinite series** (above) — impractical for large n .
2. **Laplace transform method:** $\Phi(t) = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$
3. **Cayley-Hamilton theorem:** Express $e^{\mathbf{A}t}$ as a polynomial in $\mathbf{A}$ of degree $\leq n - 1$.
4. **Diagonalisation:** If $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$, then $e^{\mathbf{A}t} = \mathbf{P}e^{\mathbf{\Lambda}t}\mathbf{P}^{-1}$.

Laplace Method (Most Exam-Relevant)

$$\Phi(t) = e^{\mathbf{A}t} = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$$

Steps:

1. Form $(s\mathbf{I} - \mathbf{A})$ by subtracting $\mathbf{A}$ from s times the identity matrix.
2. Compute the inverse: $(s\mathbf{I} - \mathbf{A})^{-1} = \frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})}$
3. Take the inverse Laplace transform of each element.

3.2 Properties of the STM

#	Property	Expression
1	Initial value	$\Phi(0) = \mathbf{I}$ (identity matrix)
2	Inverse	$\Phi^{-1}(t) = \Phi(-t) = e^{-\mathbf{A}t}$
3	Composition	$\Phi(t_1 + t_2) = \Phi(t_1) \Phi(t_2)$
4	Derivative	$\dot{\Phi}(t) = \mathbf{A} \Phi(t) = \Phi(t) \mathbf{A}$
5	Transition	$\Phi(t_2 - t_1) = \Phi(t_2) \Phi^{-1}(t_1)$
6	Determinant	$\det[\Phi(t)] = e^{\text{tr}(\mathbf{A}) \cdot t} \neq 0$ (always non-singular)

Example: Compute STM for $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$

$$(s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$\det(s\mathbf{I} - \mathbf{A}) = s(s+3) + 2 = s^2 + 3s + 2 = (s+1)(s+2)$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{(s+1)(s+2)} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

Using partial fractions and inverse Laplace:

$$\Phi(t) = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

4. Solution of State Equations

4.1 Homogeneous State Equation (Zero Input)

When $\mathbf{u}(t) = \mathbf{0}$, the state equation reduces to:

$$\dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t)$$

Solution — Homogeneous (Zero Input)

$$\mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{x}(0) = \Phi(t) \mathbf{x}(0)$$

This is the **free response** or **zero-input response**. It depends only on the initial state $\mathbf{x}(0)$ and the system matrix $\mathbf{A}$.

The output is: $\mathbf{y}(t) = \mathbf{C} \Phi(t) \mathbf{x}(0)$

Derivation: Assume $\mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{c}$. Then $\dot{\mathbf{x}} = \mathbf{A} e^{\mathbf{A}t} \mathbf{c} = \mathbf{A} \mathbf{x}$. At $t = 0$: $\mathbf{x}(0) = \mathbf{c}$. Hence $\mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{x}(0)$.

4.2 Non-Homogeneous State Equation (With Input)

When $\mathbf{u}(t) \neq \mathbf{0}$:

$$\dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t) + \mathbf{B} \mathbf{u}(t)$$

Solution — Non-Homogeneous (With Input)

$$\mathbf{x}(t) = \underbrace{e^{\mathbf{A}t} \mathbf{x}(0)}_{\text{zero-input}} + \underbrace{\int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B} \mathbf{u}(\tau) d\tau}_{\text{zero-state (convolution)}}$$

$$= \Phi(t) \mathbf{x}(0) + \int_0^t \Phi(t - \tau) \mathbf{B} \mathbf{u}(\tau) d\tau$$

Output:

$$\mathbf{y}(t) = \mathbf{C} \Phi(t) \mathbf{x}(0) + \int_0^t \mathbf{C} \Phi(t - \tau) \mathbf{B} \mathbf{u}(\tau) d\tau + \mathbf{D} \mathbf{u}(t)$$

Derivation (variation of parameters): Multiply both sides of $\dot{\mathbf{x}} - \mathbf{A}\mathbf{x} = \mathbf{B}\mathbf{u}$ by the integrating factor $e^{-\mathbf{A}t}$:

$$\frac{d}{dt} [e^{-\mathbf{A}t} \mathbf{x}(t)] = e^{-\mathbf{A}t} \mathbf{B} \mathbf{u}(t)$$

Integrating from 0 to t and multiplying by $e^{\mathbf{A}t}$ gives the result above.

4.3 Laplace Domain Solution

Taking the Laplace transform of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$:

Laplace Domain Solution

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s)$$

$$(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{x}(0) + \mathbf{B}\mathbf{U}(s)$$

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B}\mathbf{U}(s)$$

$$\mathbf{Y}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{x}(0) + [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} + \mathbf{D}] \mathbf{U}(s)$$

5. Transfer Function from State Space

The **transfer function matrix** $\mathbf{G}(s)$ relates the output $\mathbf{Y}(s)$ to the input $\mathbf{U}(s)$ with zero initial conditions $[\mathbf{x}(0) = \mathbf{0}]$:

$$\mathbf{G}(s) = \left. \frac{\mathbf{Y}(s)}{\mathbf{U}(s)} \right|_{\mathbf{x}(0)=\mathbf{0}}$$

Transfer Function from State Space (Zero I.C.)

Setting $\mathbf{x}(0) = \mathbf{0}$ in the Laplace solution:

$$\mathbf{Y}(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}] \mathbf{U}(s)$$

$$\boxed{G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}}$$

For SISO systems ($\mathbf{D} = 0$ typically): $G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$

The **characteristic polynomial** is $\det(s\mathbf{I} - \mathbf{A})$, whose roots are the **eigenvalues** of $\mathbf{A} =$ poles of the system.

Example: TF from State Space

Given: $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $D = 0$

$$(s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}, \quad \det = s^2 + 3s + 2$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

$$G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{s^2 + 3s + 2}$$

Exam Tips — State Space & STM

- **STM** = $e^{\mathbf{A}t} = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$ — Laplace method is fastest in exams.
- **Homogeneous solution:** $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0)$ — zero input, only initial state.
- **Non-homogeneous solution:** zero-input + zero-state (convolution integral).
- **TF from SS:** $G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$ — set $\mathbf{x}(0) = \mathbf{0}$.
- **Poles** of the system = eigenvalues of $\mathbf{A}$ = roots of $\det(s\mathbf{I} - \mathbf{A}) = 0$.
- $\Phi(0) = \mathbf{I}$ always — a quick check to verify your STM computation.

6. Controllability

Controllability: A system is said to be **completely state controllable** if it is possible to transfer the system from *any* initial state $\mathbf{x}(t_0)$ to *any* desired final state $\mathbf{x}(t_f)$ in a finite time $t_f - t_0$ using an unconstrained control input $\mathbf{u}(t)$.

In simple terms: *Can the input reach and control every state variable?*

6.1 Kalman's Controllability Test

Controllability Matrix

The system $(\mathbf{A}, \mathbf{B})$ is **completely controllable** if and only if the **controllability matrix** $\mathbf{Q}_c$ has full rank n :

$$\mathbf{Q}_c = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B} \quad \dots \quad \mathbf{A}^{n-1}\mathbf{B}]$$

$$\text{rank}(\mathbf{Q}_c) = n \Leftrightarrow \det(\mathbf{Q}_c) \neq 0 \Leftrightarrow \text{System is controllable}$$

$\mathbf{Q}_c$ is an $n \times nr$ matrix (for SISO: $n \times n$ square matrix).

Example: Check Controllability

Given: $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $n = 2$

$$\mathbf{AB} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$\mathbf{Q}_c = [\mathbf{B} \quad \mathbf{AB}] = \begin{bmatrix} 0 & 1 \\ 1 & -3 \end{bmatrix}$$

$$\det(\mathbf{Q}_c) = (0)(-3) - (1)(1) = -1 \neq 0$$

Result: $\text{rank}(\mathbf{Q}_c) = 2 = n \Rightarrow$ System is **completely controllable**.

6.2 Physical Interpretation

- If a system is **not controllable**, some state variables are *decoupled* from the input — no matter what $\mathbf{u}(t)$ is applied, those states cannot be driven to a desired value.
- **Pole placement** (assigning closed-loop poles arbitrarily) is possible *only if* the system is completely controllable.
- An uncontrollable mode corresponds to a pole of the system that cannot be moved by state feedback.

7. Observability

Observability: A system is said to be **completely observable** if every state variable $x_i(t_0)$ can be determined from the output $\mathbf{y}(t)$ over a finite time interval $[t_0, t_f]$, given the input $\mathbf{u}(t)$.

In simple terms: *Can we reconstruct every state variable by observing the output?*

7.1 Kalman's Observability Test

Observability Matrix

The system $(\mathbf{A}, \mathbf{C})$ is **completely observable** if and only if the **observability matrix** $\mathbf{Q}_o$ has full rank n :

$$\mathbf{Q}_o = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \mathbf{CA}^2 \\ \vdots \\ \mathbf{CA}^{n-1} \end{bmatrix}$$

$$\text{rank}(\mathbf{Q}_o) = n \Leftrightarrow \det(\mathbf{Q}_o) \neq 0 \Leftrightarrow \text{System is observable}$$

$\mathbf{Q}_o$ is an $mn \times n$ matrix (for SISO: $n \times n$ square matrix).

Example: Check Observability

Given: $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}, n = 2$

$$\mathbf{CA} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$\mathbf{Q}_o = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\det(\mathbf{Q}_o) = 1 \neq 0$$

Result: $\text{rank}(\mathbf{Q}_o) = 2 = n \Rightarrow$ System is **completely observable**.

7.2 Physical Interpretation

- If a system is **not observable**, some state variables are *hidden* from the output — they cannot be estimated from measurements.
- **State observer (Luenberger observer)** design is possible *only if* the system is completely observable.
- An unobservable mode corresponds to a pole that does not appear in the transfer function (pole-zero cancellation).

7.3 Controllability vs Observability — Duality

Duality Principle (Kalman)

The system $(\mathbf{A}, \mathbf{B}, \mathbf{C})$ is **controllable** if and only if the **dual system** $(\mathbf{A}^T, \mathbf{C}^T, \mathbf{B}^T)$ is **observable**, and vice versa.

This means: $\mathbf{Q}_c(\mathbf{A}, \mathbf{B}) = \mathbf{Q}_o^T(\mathbf{A}^T, \mathbf{C}^T)$

Property	Controllability	Observability
Definition	Transfer any state to any other	Determine any state from output
Test matrix	$\mathbf{Q}_c = [\mathbf{B} \ \mathbf{A}\mathbf{B} \ \dots \ \mathbf{A}^{n-1}\mathbf{B}]$	$\mathbf{Q}_o = [\mathbf{C}^T \ (\mathbf{C}\mathbf{A})^T \ \dots]^T$
Condition	$\text{rank}(\mathbf{Q}_c) = n$	$\text{rank}(\mathbf{Q}_o) = n$
Matrices involved	$\mathbf{A}$ and $\mathbf{B}$	$\mathbf{A}$ and $\mathbf{C}$
Application	Pole placement, state feedback	State observer (Luenberger)
Dual of	Observability of $(\mathbf{A}^T, \mathbf{C}^T)$	Controllability of $(\mathbf{A}^T, \mathbf{C}^T)$

Exam Tips — Controllability & Observability

- **Controllability matrix:** $\mathbf{Q}_c = [\mathbf{B} \ \mathbf{A}\mathbf{B} \ \mathbf{A}^2\mathbf{B} \ \dots \ \mathbf{A}^{n-1}\mathbf{B}]$ — involves $\mathbf{A}$ and $\mathbf{B}$.
- **Observability matrix:** $\mathbf{Q}_o = [\mathbf{C}^T \ (\mathbf{C}\mathbf{A})^T \ \dots \ (\mathbf{C}\mathbf{A}^{n-1})^T]^T$ — involves $\mathbf{A}$ and $\mathbf{C}$.
- Both tests: compute the matrix, find its determinant. Non-zero $\Rightarrow$ full rank $\Rightarrow$ controllable/observable.
- **Pole-zero cancellation** in TF $\Rightarrow$ system is either uncontrollable or unobservable (or both).
- **Duality:** $(\mathbf{A}, \mathbf{B})$ controllable $\Leftrightarrow (\mathbf{A}^T, \mathbf{B}^T)$ observable.
- For a system in **phase variable form**: always controllable; observable depends on $\mathbf{C}$.

Quick Revision — Module 5

State Variable Analysis | EC601

Key Formulas at a Glance

State equation	$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$
Output equation	$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$
STM definition	$\Phi(t) = e^{\mathbf{A}t}$
STM via Laplace	$\Phi(t) = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$
STM series	$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2 t^2}{2!} + \dots$
Homogeneous solution	$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0)$
Non-homogeneous solution	$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau$
Transfer function from SS	$G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$
Characteristic polynomial	$\det(s\mathbf{I} - \mathbf{A}) = 0$
Controllability matrix	$\mathbf{Q}_c = [\mathbf{B} \ \mathbf{A}\mathbf{B} \ \dots \ \mathbf{A}^{n-1}\mathbf{B}]$
Controllable condition	$\text{rank}(\mathbf{Q}_c) = n$
Observability matrix	$\mathbf{Q}_o = [\mathbf{C}^T \ (\mathbf{C}\mathbf{A})^T \ \dots \ (\mathbf{C}\mathbf{A}^{n-1})^T]^T$
Observable condition	$\text{rank}(\mathbf{Q}_o) = n$
STM property	$\Phi(0) = \mathbf{I}; \Phi^{-1}(t) = \Phi(-t)$

One-Line Definitions

- **State:** Minimum set of variables that, with input, completely determines future system behaviour.
- **State Variable:** Each member of the minimum set; n variables for an n -th order system.
- **State Vector $\mathbf{x}$:** Column vector of all n state variables.
- **State Space:** n -dimensional space with state variables as axes.
- **A matrix:** System/plant matrix ($n \times n$); its eigenvalues are the system poles.
- **B matrix:** Input matrix ($n \times r$); couples input to state derivatives.
- **C matrix:** Output matrix ($m \times n$); selects which states appear in output.
- **D matrix:** Direct transmission matrix; usually zero for physical systems.
- **STM $\Phi(t)$:** Matrix exponential $e^{\mathbf{A}t}$; maps initial state to state at time t .
- **Homogeneous solution:** Zero-input response; $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0)$.

- **Non-homogeneous solution:** Zero-input + zero-state (convolution with input).
- **Controllability:** Ability to drive any state to any desired value using input.
- **Observability:** Ability to determine any state from output measurements.
- **Duality:** $(\mathbf{A}, \mathbf{B})$ controllable $\Leftrightarrow (\mathbf{A}^T, \mathbf{C}^T)$ observable.

Frequently Asked / Viva Questions

Q1. What is a state variable?

The minimum set of variables that, together with the input, completely determines the future behaviour of the system. An n -th order system has exactly n state variables.

Q2. What is the State Transition Matrix?

$\Phi(t) = e^{\mathbf{A}t}$. It maps the initial state to the state at time t in the absence of input: $\mathbf{x}(t) = \Phi(t)\mathbf{x}(0)$.

Q3. How is the STM computed using Laplace transform?

$\Phi(t) = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$. Form $(s\mathbf{I} - \mathbf{A})$, invert it, then take the inverse Laplace of each element.

Q4. What is the solution of the homogeneous state equation?

$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0)$. This is the zero-input (free) response depending only on the initial state.

Q5. What is the solution of the non-homogeneous state equation?

$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau$. Sum of zero-input and zero-state (convolution) responses.

Q6. How is the transfer function obtained from state space?

Set $\mathbf{x}(0) = \mathbf{0}$ and take Laplace transform: $G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$.

Q7. What are the poles of a state space system?

The eigenvalues of the $\mathbf{A}$ matrix, i.e., roots of $\det(s\mathbf{I} - \mathbf{A}) = 0$ (characteristic equation).

Q8. Define controllability and state its test.

A system is controllable if any state can be driven to any desired state in finite time. Test: $\text{rank}(\mathbf{Q}_c) = n$ where $\mathbf{Q}_c = [\mathbf{B} \ \mathbf{A}\mathbf{B} \ \dots \ \mathbf{A}^{n-1}\mathbf{B}]$.

Q9. Define observability and state its test.

A system is observable if every state can be determined from the output in finite time. Test: $\text{rank}(\mathbf{Q}_o) = n$ where $\mathbf{Q}_o = [\mathbf{C}^T \ (\mathbf{C}\mathbf{A})^T \ \dots]^T$.

Q10. What is the duality principle?

$(\mathbf{A}, \mathbf{B})$ is controllable if and only if $(\mathbf{A}^T, \mathbf{B}^T)$ is observable. Controllability and observability are dual concepts.

Frequently Asked / Viva Questions (contd.)

Q11. What does pole-zero cancellation in TF indicate?

It indicates the system is either uncontrollable or unobservable (or both). The cancelled pole is a mode that is decoupled from either the input or the output.

Q12. What are the properties of the STM?

- $\Phi(0) = \mathbf{I}$ (identity at $t = 0$)
- $\Phi^{-1}(t) = \Phi(-t)$
- $\Phi(t_1 + t_2) = \Phi(t_1) \Phi(t_2)$
- $\dot{\Phi}(t) = \mathbf{A} \Phi(t)$

State Space Matrices — Quick Reference

Matrix	Size	Name	Role
A	$n \times n$	System matrix	Governs state dynamics; eigenvalues = poles
B	$n \times r$	Input matrix	Couples input to state derivatives
C	$m \times n$	Output matrix	Selects states visible in output
D	$m \times r$	Feedthrough matrix	Direct input-to-output path (usually 0)
$\mathbf{Q}_c$	$n \times nr$	Controllability matrix	Full rank $\Rightarrow$ controllable
$\mathbf{Q}_o$	$mn \times n$	Observability matrix	Full rank $\Rightarrow$ observable